Instructions for building and using the SCHIRTE measurement system

Contents

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1) Description, Performance Limits & Repository-structure

The SCHIRTE ([ˈʃɪrtə] SCHock-Intensitäten-Resistenz Test-Einrichtung, Engl.: SHock Intensity Resistance Test Equipment) measures accelerations or, more specifically, accelerations of vibrations. Primarily, the measurement system was developed to acquire data on brake shocks on motor shafts. For more information on brake shocks and their effects on rotary encoders, please refer to the corresponding paper "*Influence of brake shocks on safety components of electric motors; Napp/Schepers*" presented at ESREL 2026. (See internet page https://www.esrel2026.com). The recorded data can be used to justify a "proven-in-use" approach for qualifying safety-critical encoders concerning their lifetime under the influence of brake shocks.

In addition, the measurement system can be used in other environments where vibrations may occur. To achieve this, an adapted housing for the specific application must be developed, distinct from the one described in this paper.

Generally, the system consists of two PCBs. On the main PCB, a triaxial acceleration sensor is mounted to acquire the shock data (Sensor PCB). The second PCB is for the wireless data transfer via Bluetooth and for loading the LiPo battery (Bluetooth/Power PCB).

The system's performance limits are determined, among other factors, by the sensor used. The sensor has a measuring range of ±2000 g and a cutoff frequency of 10 kHz. The operating temperature range is -40 °C to 125 °C. Additionally, the sampling rate is 97.55 kHz if the channels x and z are recorded sequentially. However, if the channels are recorded simultaneously, the sampling rate is reduced to 55.42 kHz for each channel. Note: These sampling rates are the maximum which could be reached with the applied microcontroller.

The default trigger level is 50 g, but it can be changed in the software (e.g. if noise causes continuous trigger events). The measuring system consumes around 40 mA during operation. With a 400 mAh battery, the system's running time should be max. 10 h, though it is likely lower given the battery's discharge curve. 4 h – 5 h running time should be realistic. The battery can be charged via the Micro USB connector on the Bluetooth/Power PCB.

In the GitHub repository, you can find all the documents required to build and use the SCHIRTE system. This includes the KiCAD and Gerber files for manufacturing the PCBs, the source code for programming the microcontroller via UPDI, and the source code for the Open-Source software Octave for analysing and displaying the recorded data. In addition, you can find examples of datasets and plots.

2) Safety and Liability Information

The measurement system was developed to record brake shocks across various motor-brake configurations and on a test bench, in accordance with ISO 61800-5-3, Annex D.

The following paragraphs contain safety instructions for the use of LiPo batteries; this information is provided without any guarantee of completeness.

LiPo batteries have a high energy density. Improper handling can lead to fire, excessive heat, smoke, and explosion. Use LiPo batteries only if you are familiar with their handling, and observe the following precautions:

General Information:

- Never short-circuit, crush, puncture, bend, or otherwise mechanically damage LiPo batteries.
- Do not open the batteries or allow them to come into contact with water.
- Do not continue to use the battery if it is swollen, leaking, emits an unusual odour, becomes very hot, or shows visible damage.

Charging:

- Charge the proposed LiPo battery only with the charger on the provided PCB. Use alternative LiPo batteries only with a charger designed for them and the appropriate charging method (e.g., CC/CV; for multi-cell packs, use balancing).
- Never leave the device charging unattended. Charge it on a non-flammable surface and keep it a safe distance away from flammable materials.
- Make sure the cell count and voltage are correct and that you do not exceed the maximum charging rate specified by the manufacturer.
- Do not charge batteries when they are cold or hot; let them come to room temperature first.

Operation Mode/Discharging:

- Avoid deep discharge. Use an appropriate low-voltage cutoff or monitor the cell voltage.
- Prevent overcurrent (e.g., due to improperly sized cables or connectors). Use appropriate fuses or protective circuits whenever possible.
- Operate the battery only within the specified temperature range; protect it from excessive heat (e.g., direct sunlight, proximity to hot components).

Storage and Transport:

- Store LiPo batteries in a cool, dry, and non-flammable location (e.g., in a suitable LiPo safety bag or metal container, with no risk of short-circuiting).
- For long-term storage, charge the battery to storage voltage (typically approx. 3.7–3.85 V per cell; follow the manufacturer’s instructions).

- Protect contacts against short circuits; do not transport batteries together with conductive objects.

What to Do in Case of an Error:

- If smoke or heat develops: If it is safe to do so, immediately take the battery outdoors and keep a safe distance.
- Extinguishing: Use sand, metal fire extinguishing agents, or plenty of water to cool the surrounding area (depending on the situation); follow local fire department guidelines. Batteries may reignite.
- After an incident, do not continue to use the battery; dispose of it properly (at a collection point).

Note: Use of this measurement system with a LiPo battery is at your own risk. It is assumed that appropriate safety measures (e.g., charge/discharge protection, proper wiring, and mechanical protection of the battery) have been implemented.

Disclaimer: SCHIRTE is a prototype measuring system. Use at your own risk. The authors provide this hardware and software as-is, with no warranties or guarantees; we accept no liability for any damage, injury, or loss arising from its use.

3) Requirements

Hardware manufacturing:

The components for PCB manufacturing are almost exclusively surface-mount components. Only the connector header for the Bluetooth module is a through-hole solder joint. Using the Gerber files provided in the KiCAD project, the PCBs can (should) be manufactured and assembled by a PCB manufacturer.

Additional materials required for the hardware include cables for programming the microcontroller via the UPDI interface and for connecting the two boards. For on-chip programming of the microcontroller, the Atmel-ICE programming interface (or an appropriate alternative) must be used.

For the housing, the upper housing section can be manufactured using a 3D printer. A material should be used that ensures the Bluetooth signal can pass through. The lower housing section can be made of aluminium, which requires appropriate machinery and tools for machining aluminium. To mount the housing onto the motor shaft, a grub screw is required, along with M3 screws to assemble the two housing parts and screw connectors to connect the two circuit boards.

Additional required hardware:

- Atmel-ICE programming interface

Required software:

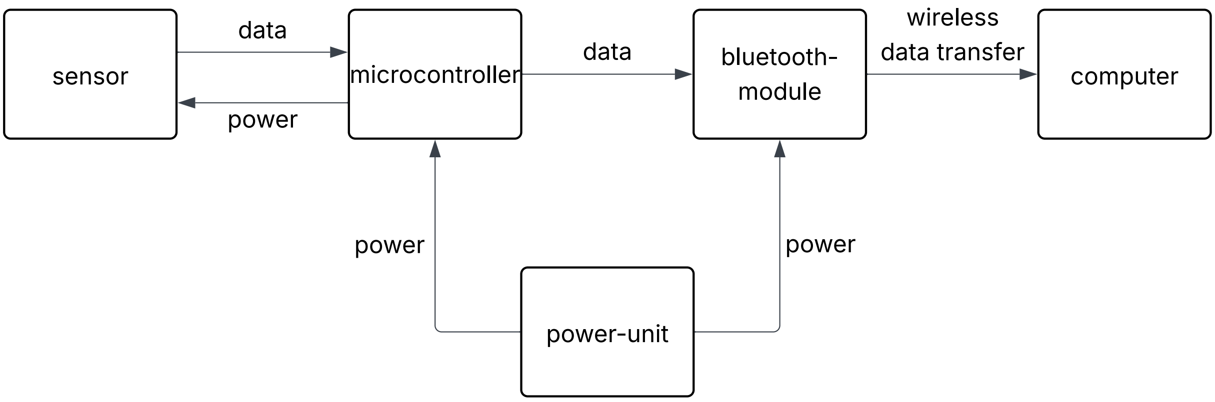
- KiCAD at least version 9.0
- MPLAB X IDE for microcontroller programming at least version 6.15
- Octave at least version 10.3.0 for data analysis software
- hTerm for receiving the Bluetooth data from the measuring system (at least version 0.8.9) download: https://www.der-hammer.info/pages/terminal.html
- Editor program for saving the received data in a txt-file (default editor program on Windows or preferred one)

4) Repository structure

- /hardware/SCHIRTE_BT-Power-PCB_2-0_KiCad_Gerber/, /hardware/SCHIRTE_Sensor-PCB_3-1_KiCad_Gerber_Footprints/
- /firmware/ (source code, Releases)
- /software_analysis/ (code files)
- /docs/ (step-by-step manual for analysing software)
- /examples/ (example data, example diagrams of shocks and PVSRS)
- license in /readme.md

5) Hardware

System overview:



The measurement system consists primarily of two circuit boards. The first circuit board is responsible for data acquisition and processing (Sensor PCB). The microcontroller prepares the data for transmission to a computer via the Bluetooth module. This circuit board also houses the UPDI interface for programming the microcontroller.

The second circuit board contains components for Bluetooth data transmission and the hardware for charging the connected battery (Bluetooth/Power PCB). The Bluetooth module used is a standard HC-05.

Bill of Materials:

Sensor PCB

Labels in KiCAD Project	Description (manufacturer)	Part Number	Quantity
C1, C2, C3, C4, C5, C6	Capacitor (0805); 0,1 µF		6
C7, C8	Capacitor (0805); 1 µF		2
R1, R2, R3	Resistor (0805); 100 Ohm		3
R4	Resistor (0805); 10 kOhm		1
U3	Voltage regulator (Microchip)	MCP1700T-3302E/TT	1
SW1	DIP switch (Omron)	A6S-1101-H	1
U1	Operational amplifier (Microchip)	MCP604-I/SL	1
U2	Triaxial acceleration sensor (TE connectivity)	830M1-2000	1
U5	Microcontroller (Microship)	AVR128DA28-I/SS	1
Y1	Oscillator (Würth electronics)	LFSPXO019082	1

- J1: Connection pad for power input (from Bluetooth/Power PCB)
- J2: Connection pad for ground (from Bluetooth/Power PCB)
- J3: Pad for connection between TxD (Sensor PCB) and RxD (Bluetooth/Power PCB)
- J4: Pad for connection between RxD (Sensor PCB) and TxD (Bluetooth/Power PCB)
- J5: Pad for 3.3 V output for UPDI-programming
- J6: Pad for ground output for UPDI-programming
- J7: Pad for data connection of UPDI-programming

Bluetooth/Power PCB

Labels in the KiCAD project	Description (manufacturer)	Part number	Quantity
C2	Capacitor (0805); 4,7 µF		2
R1	Resistor (0805); 470 Ohm		1
R2	Resistor (0805); 2 kOhm		1
D1	LED red (0805) (Everlight)	EVL-17-21USRC	1
J4	MicroUSB Port (Würth electronics)	629105150521	1
J9	Female header (Fischer Elektronik)	FIS BL312G	1
S1	Power switch (Diptronics)	LSSA22L-V	1
U1	Battery charger (Microchip)	MCP73831T-2ACI/OT	1

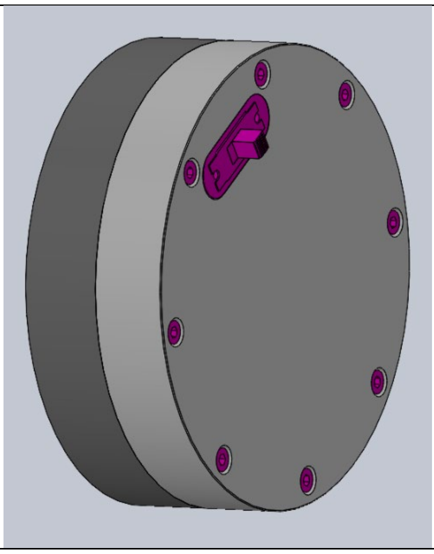
- J5: Power output for connection to Sensor PCB
- J6: Ground output for connection to Sensor PCB
- J1: Plus (+) input from LiPo battery
- J7: Minus (-) input from LiPo battery
- J2: Pad for connection between RxD (Bluetooth/Power PCB) and TxD (Sensor PCB)
- J3: Pad for connection between TxD (Bluetooth/Power PCB) and RxD (Sensor PCB)

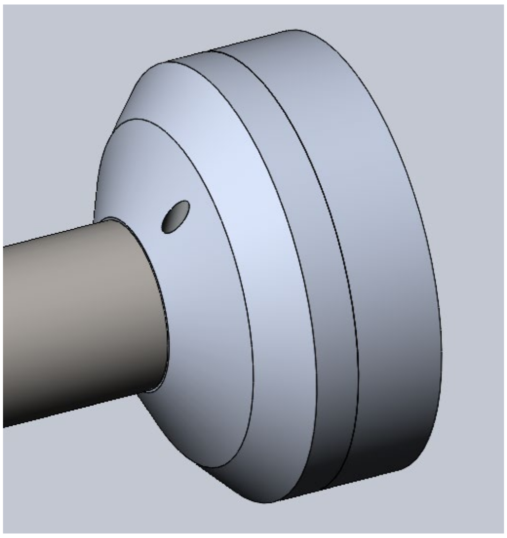
PCB parameters:

Both circuit boards are four-layer and measure 40 x 40 mm. Depending on the PCB manufacturer, components for assembly can either be ordered by the customer and delivered for assembly or ordered by the manufacturer. The production files for the PCBs can be found in the repository at:

Solder resist	green
Solder surface	unleaded
Trace width	0.300 mm (outer layers); 0.250 mm (inner layers)
Isolation distance	0.200 mm (outer layer); 0.250 mm (inner layer)
Outer layer restriction	0.125 mm
Inner layer restriction	0.125 mm
Drilling grade	C
Material thickness	1.55 mm
Material type	FR4IMP
Outer layers of copper foil	18 start-/ 35 end copper
Material Tg	145-150°C
Inner layers of copper foil	35 µm

Housing (proposal):





The housing described here is intended for use on motor shafts only. As shown in the illustrations, the housing should be round to minimise vibration when the shaft rotates. It is also recommended that the housing be manufactured in two parts. The lower part, in which the data acquisition circuit board is mounted, should be made of aluminium to protect against electromagnetic interference. The upper part should be made of plastic, as this is where the Bluetooth module is located, and plastic ensures permeability to the Bluetooth signal.

A recess should be cut out in the lower part of the housing to match the diameter of the shaft onto which the housing is to be mounted. The housing is secured to this shaft using a grub screw. Furthermore, when designing a housing, care must be taken to ensure that the underside of the housing rests flat on the surface to be measured.

Very important note: A non-damping insulating layer must be used between the metal housing and the underside of the Sensor PCB to prevent short circuits caused by exposed copper via (through-hole) contacts! Further information and proposals for mounting the Sensor PCB can be found in the datasheet of the 830M1-2000 triaxial sensor.

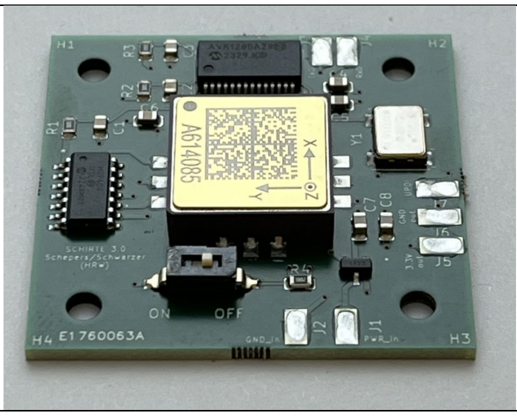
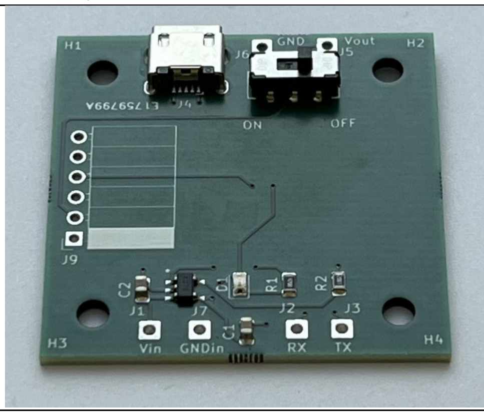
6) Commissioning

First, the battery must be soldered to the soldering pads on the Bluetooth/Power PCB. Via the Atmel ICE device and the UPDI interface, the microcontroller can be programmed (see below). The battery plus pole must be connected to J1 (Vin) and the minus pole to J7 (GNDin) of the Bluetooth/Power PCB.

Important note: The power switch on the Bluetooth/Power PCB should be OFF when soldering the battery to the PCB.

When the battery is connected, it can be charged via the USB connector. On the PCB of the Bluetooth/Power module, there is a red LED which indicates battery charging.

Then the two PCBs must be electrically connected with cables. The applied cables should not be too short to allow stacking of the PCBs for final mounting in the housing. The cables must be soldered to the PCBs. The sensor PCB provides contact pads (no through-holes) because otherwise it could not be mounted flat on the bottom of the housing.

Sensor PCB	Bluetooth/Power PCB
	

The following pads must be connected:

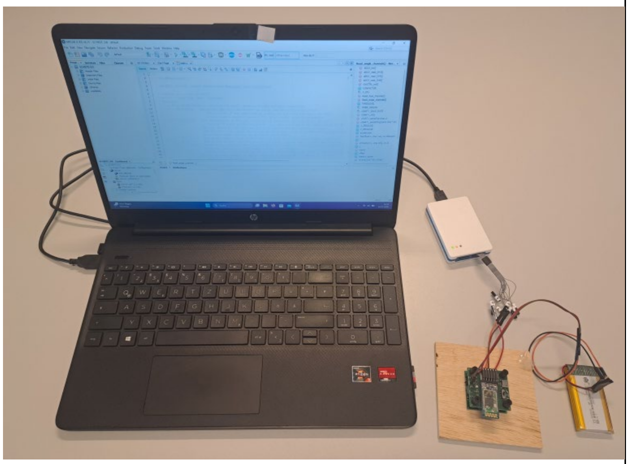
Sensor PCB		*Bluetooth/Power PCB*
J1 (Power_in)	↔	J5 (Vout)
J2 (GND_in)	↔	J6 (GND)
J3 (TxD)	↔	J2 (RX)
J4 (RxD)	↔	J3 (TX)

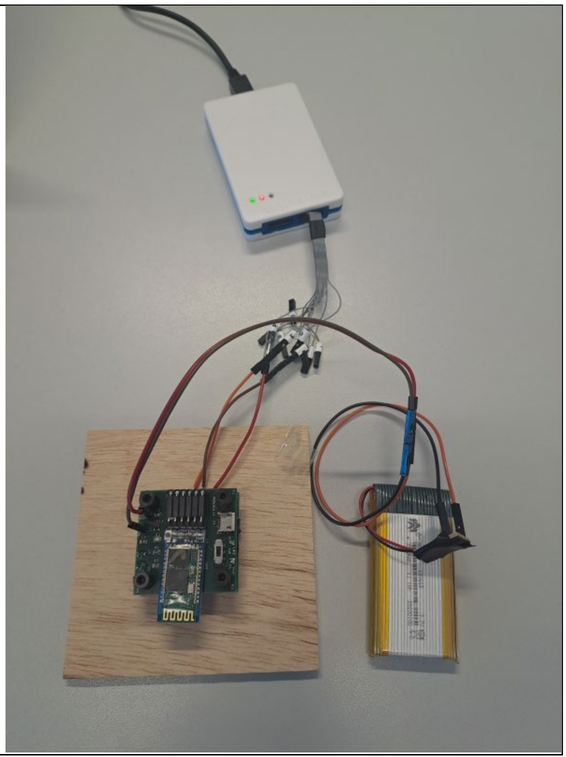
Finally, the UPDI interface must be connected. For this purpose, cables must be soldered to the sensor PCB. The cables should have male connectors on one side so they can be easily connected to the Atmel-ICE programming interface. In total, three cables must be soldered to the connectors J5 (3.3V out), J6 (GND out), and J7 (UPDI). Note: After programming, these three cables may be removed again as they serve only for programming. While prototyping, it makes sense to keep the cables in place, as reprogramming might be necessary (e.g., to modify the trigger level).

Programming of the sensor PCB:

The sensor PCB must be connected to the Atmel-ICE (or other appropriate) programming interface. For Atmel-ICE, the following connections must be established via the 10-pin cable provided with the Atmel-ICE (refer to the numbered cables of the Atmel-ICE interface):

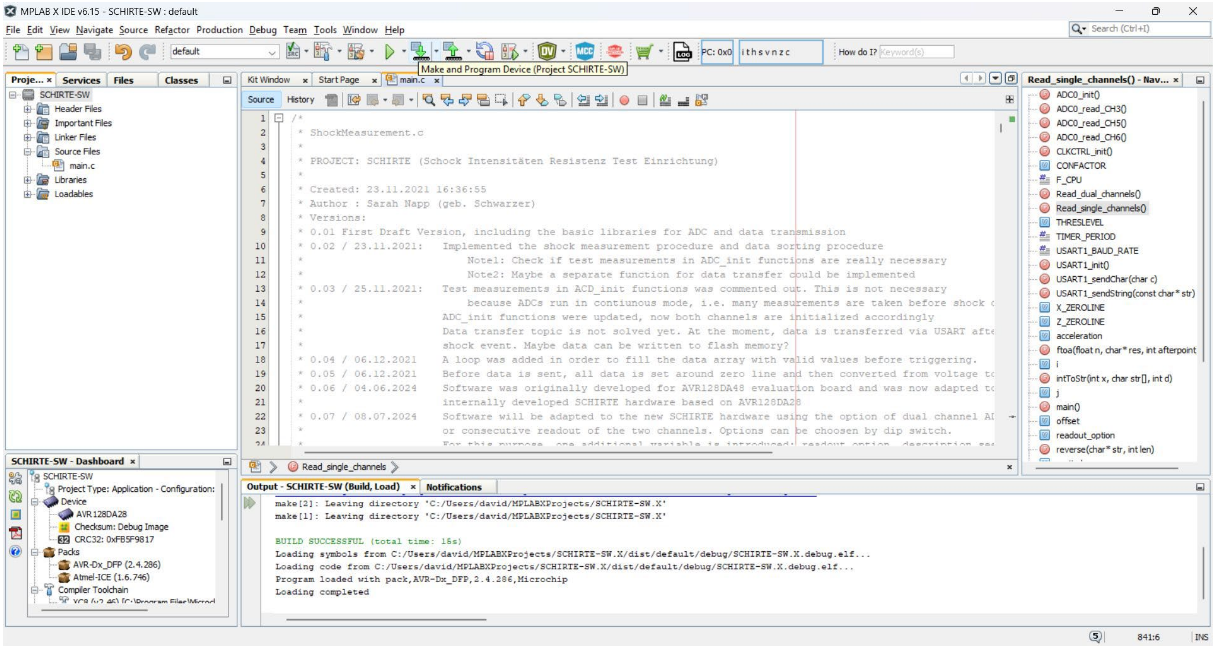
Sensor PCB		*Atmel-ICE interface*
J5 (3.3V out)	↔	Cable No. 4
J6 (GND out)	↔	Cable No. 2
J7 (UPDI)	↔	Cable No. 3



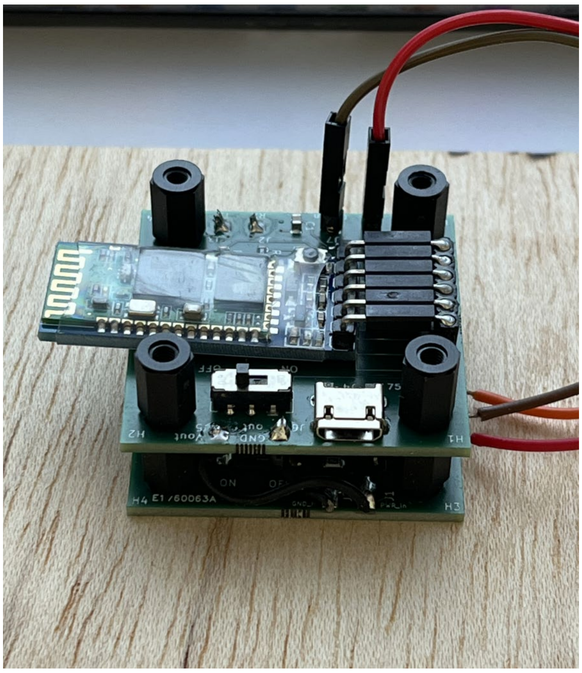


The 10-pin cable must be connected to the AVR plug of the Atmel-ICE programming interface. For programming, the Atmel MPLAB X IDE software is used. For this purpose, the MPLAB X project must be unzipped (2026-05-27_SCHIRTE-SW.X_V1.zip) within the “MPLABXProjects” folder (depending on the specific MPLAB X installation). Then the project folder may be opened in the MPLAB X IDE (“Open Project” -> Folder “SCHIRTE-SW.X”). The project contains all necessary settings and the source code. If necessary, the “main.c” file should be opened by double-clicking it (in the project file overview on the left side). The source code contains many comments for better understanding. Normally, it should not be necessary to modify the code, except for the trigger level. The standard trigger level is set to 50 g. If the measurement is affected by high noise (due to electromagnetic interference), it may be necessary to set a higher threshold to avoid continuous triggering. Note: Electromagnetic interference should be avoided for good data quality. Appropriate housing for the Sensor PCB (e.g., an aluminium housing) can shield the sensor from electromagnetic interference.

When the sensor PCB is connected to the Atmel-ICE programming interface, the interface is connected to the PC via USB, and the power switch on the sensor PCB is in the “ON” position, programming can be done by clicking on the “Make & Program Device” icon in the top toolbar of the MPLAB X software. If programming is successful, a message “Loading complete” is displayed in the output window.



For prototyping, the PCB may be mounted to a (wooden) plate. You may test the measuring system by carefully hitting an item (e.g., a small hammer) onto the plate.



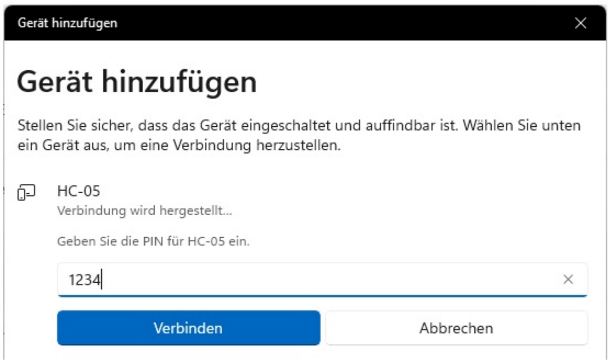
The Bluetooth module HC-05 can be connected via a 90° connector (see above) and fixed to the PCB (e.g., with double-sided adhesive tape), or the module's connector can be unsoldered so that the HC-05 can be mounted directly to the PCB.

Note: Unsoldering the HC-05 connector is not easy. The above solution is the easiest. It also has the advantage that the antenna stands out slightly from the PCB. Thus, for good Bluetooth transmission, the antenna could be kept outside the housing. The basic intention was to keep the measurement system simple; therefore, the standard HC-05 module was used. In future versions, the PCB could be modified, and an SMD Bluetooth module might be mounted directly on it.

Important note: If the system does not trigger, this might be due to the alignment of the trigger axis. If the X-axis triggers first, the plate must be hit in this direction (e.g., on the edge of the plate, not on the bottom). Hitting the bottom corresponds to a shock along the Z-axis.

Data acquisition can be set to parallel (55.42 kHz; set the switch on the sensor PCB to “ON”) or sequential (97.55 kHz; set the switch on the sensor PCB to “OFF”).

For data transfer between the measuring system and a PC, hTerm must be installed on the PC. The Bluetooth connection between the PC and the HC-05 module must be established (on Windows via Settings > Bluetooth > Connect new device). In most cases, HC-05 requires a password. Most HC-05 modules have the default password “1234”. In hTerm, the correct COM port must be selected, and the Bluetooth module's Baud Rate set to 115200 baud.

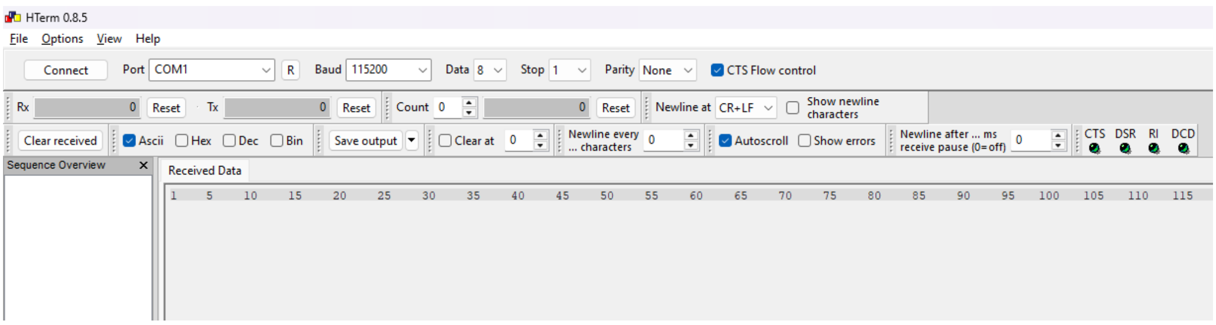


(Sorry, German Windows OS …)

Finally, the “CTS flow” box must be ticked, and it is recommended to set New Line to “CR + LF” and untick “Show newline characters”. Then, a connection to the HC-05 module can be established by clicking on “Connect”.

Note: “Trial & Error” might be necessary to find the correct COM port for the Bluetooth connection, as the active Bluetooth connection may not indicate which COM port it uses.

Note: The red LED on the HC-05 modules blinks rapidly if no connection is established (after the SCHIRTE measurement system is turned on). With a successful connection to the PC, the HC-05 LED blinks slowly. If hTerm finally connects successfully to the HC-05, the red LED blinks twice, then pauses.



After the first functional tests, the PCBs should be mounted in the housing, and the whole system should be connected to the motor shaft.

7) Configuration and operation

Before operating the battery, it should be charged completely. Charging can be done using a USB (Micro USB) cable. While charging, the power switch on the Bluetooth/Power PCB should be in the “OFF” position. After the battery has finished charging, the red LED turns off, and the USB cable can be disconnected. The measuring system can be turned on by sliding the power switch into the “ON” position.

The sensor continuously measures vibration acceleration. As soon as the pre-set trigger level is exceeded, data is recorded and transmitted for 17 ms before the trigger and 43 ms after the trigger. This results in a total recording duration of approximately 60 ms. The default trigger is set to a 50 g acceleration and can be changed in the microcontroller's software.

As mentioned before, the system has two recording modes: In one mode, the x and z axes are recorded sequentially at a sampling rate of 97.55 kHz. In the other mode, both axes are sampled simultaneously, resulting in a lower sampling rate of 55.42 kHz.

When recording brake shocks, the sequential mode is recommended, as the data is applied to calculate the PVSRS using the analysis software. It is recommended that the sampling rate be ten times the maximum recorded frequency for PVSRS calculation; therefore, for a frequency spectrum up to 10 kHz, only the sequential mode fully meets this requirement.

In addition to the trigger level, the recording duration and the associated sampling rate can also be adjusted in the software.

Important note: To ensure accurate results, it is strongly recommended that you do not alter the recording duration or the sampling rate.

The transmitted data is displayed in ASCII format in the hTerm terminal, with each new transmission beginning with the message “Trigger! Data will be transferred.” The ASCII data must now be manually copied and pasted into a text file in the text editor and saved. Spaces separate the data points, and the decimal separator is “.” If any changes are made to this, adjustments must be made in the analysis software. The txt file can then be selected in the analysis software, where the PVSRS is automatically calculated and displayed in a 4CP.

8) Analysis of the data using Octave software

The provided code files can be run via the Octave GUI (download from https://octave.org/download). All code files must be located in a single directory for the analysis to run, as the main function calls upon the sub-functions. The code is designed for use on the Windows operating system.

As an alternative to a text file, a CSV file can be imported, provided it contains both the time and a voltage value in a single column. A semicolon must separate these two values, as this is the default setting for CSV files in the software. If other delimiters are used, the software must be adjusted. The CSV file format is typically used when saving data on oscilloscopes. Consequently, the software can also be used to analyse shocks on test benches that are recorded by other sensors via an oscilloscope.

To analyse the measurement data from the measurement system described in this paper, a time vector is generated in the software and appended to the measurement data.

Examples of data files and PVSRS diagrams are available at /examples.

9) Troubleshooting

If the hTerm terminal program receives no data, there may be several reasons. First, you should check whether an active Bluetooth connection has been established with the PC. You should also check that the correct COM port has been selected.

If problems cannot be solved, refer to section 13.

10) Documentation of the interfaces

The measurement system is connected to a rechargeable battery that supplies the system with 3.7 V. The battery can be charged via a micro-USB port on the second circuit board. The system also features two switches. The switch on the second circuit board turns the system on and off, whilst the DIP switch on the main circuit board is used to change the acquisition modes.

11) Licence

The measurement system and the analysis software are licensed under the Creative Commons CC-BY-NC-SA licence. Further information can be found in the Readme file.

12) Technical data of the measurement system

Supply Voltage	3.7 V (via LiPo Battery)
Shock Measuring Range	+- 2000 g
Shock Frequency Range	Up to 10 kHz in sequential data acquisition mode. Up to 5 kHz in sequential data acquisition mode.
Maximum expanded measurement uncertainty	124.2 g (approx. 3 % of measurement range) (Calculation based on GUM (Guide to the expression of uncertainty in measurement))
Operating Temperature Range	-40 °C and 125 °C
Default Trigger Level for Shock Acquisition	50 g (adjustable via software, re-programming of microcontroller via UPDI necessary)

13) Support/Contact

Consulting services may be offered on demand. Please contact:

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